Kepler's Third Law and the Orbital-to-Radial Flow Hypothesis

Corotation, Continuity and Moving Boundaries

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Abstract

The author proposes an inward-moving non-material medium that moves between and around the impermeable atoms of material bodies and can carry them with its flow. A common body–medium coupling is hypothesized to underlie weight and inertial response, without specifying a viscous drag law. This study examines the proposed orbital-to-radial transition in a specified reference frame. We separate angular corotation, equality of radial and tangential linear speeds, and vanishing tangential slip against the rotating body. They are distinct conditions. Catalogue parameters for the Sun, eight planets and the Moon determine formal corotation radii without separately adjusted constants. The known placement of Saturn's synchronous radius inside its B ring is reproduced, and Uranus has nearly circular rings well inside its calculated radius. Consequently, corotation cannot by itself be identified with compulsory radial infall. The two satellites with measured slow secular drift lie on opposite sides of their corotation radii and drift in the corresponding directions: the Moon outside recedes, Phobos inside approaches. A divergence-free three-dimensional velocity field is constructed with net inward flux, a radial component containing an $r^{-3/2}$ contribution, and a regular meridional component directed toward the equator for one sign of the coefficient. Continuity fixes the meridional coefficient, while all-direction inflow restricts the strength of the cubic radial tendency. Meridional transport can locally cancel the spatial change of the candidate invariant before outward flow appears. Maintaining a single no-slip surface rotation rate also fixes a particular initial rotational slope. These are conditional local results, not a complete evolution law. The exact invariant $u_r^2 r^3$ is not obtained throughout this field. We also derive the radial material acceleration and a corrected moving-cavity first integral, distinguishing a velocity field sampled at one time from the motion of its boundary. These results refine the hydrodynamic interpretation of the earlier preprint without claiming an observed non-material fluid or a measured universal transition radius. They supply explicit geometric trends, observational exclusions and a framework for testing smoother transition laws. An additional case study identifies a locally near-constant $u_r^2 R^3$ in a published parametric description of HD142527. Its variation is at most 1.256% over 30–35 au. This is a property of that profile, not an observational fit error. Checks of archived CO spectra identify resolution and line-profile limitations that must be included in a future independent kinematic test.

Keywords: corotation, radial inflow, planetary rings, meridional transport, incompressibility, cavity collapse.

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1 Introduction and relation to the earlier preprints

The proposed non-material medium, to which the author attributes properties of an ideal incompressible fluid or gas with zero internal viscosity, is called the E-field in the companion study [4].

Main results. The study calculates corotation radii for ten catalogued bodies and constructs a regular three-dimensional incompressible field with a cubic radial contribution. It distinguishes spatial velocity, material acceleration and flux through a moving boundary. The HD142527 comparison quantifies a local cubic tendency in a published velocity profile and uses archived CO spectra to assess the conditions for an independent observational test.

The earlier orbital studies use $E = \omega^2 R^3$ for circular motion [1, 2]. The earlier radial-flow study considers analogies [3]. The present study examines a specified case of the proposed connection between them. The author's intended radial variable is a *linear* speed. We consequently use u_r for the radial velocity and $C_r = u_r^2 r^3$ for its proposed cubic invariant. An angular frequency will not be relabelled as a linear speed to force a match.

The parameters of selected celestial sources and controlled combination of their fields are examined in the companion multiple-system study [5]. Its body-motion calculation and the medium diagnostics here address different parts of the same question. Connecting them requires an explicit coupling law, and agreement of orbital equations alone does not establish the medium mechanism.

The physical hypothesis connects orbital circulation to inward motion in a non-material medium that moves between and around the impermeable atoms of material bodies and can carry those atoms with its flow. The working hypothesis distinguishes an internally frictionless non-material medium from its coupling to material bodies, interpreted here as assemblies of atomic cavities. The author interprets the difference between the cumulative flow kinematics and the body's falling motion as an indication of that coupling, proposed to underlie both weight and inertial response. Interactions between atoms are a separate question. The quantitative coupling law remains to be derived. Here "friction" refers to the interaction between the flow and atoms rather than a specified viscous drag law. This study tests the geometry and continuity requirements of the proposed connection. A companion study treats the response of a body to the candidate flow [6], and the author's general model of matter and medium is stated separately [4]. A connected mechanism must account for geometry, body response and rotation together. The conditional results below can be assessed independently of whether the proposed physical interpretation is ultimately confirmed.

Rotating flows with radial transport have prior literature, for example the rotating porous-cylinder problem of Ilin and Morgulis [9]. Its geometry differs from the spherical construction here and supplies no direct astronomical validation. The cavity analogy likewise has a specific moving-boundary domain.

The companion study of external flow around impermeable atoms [23] distinguishes pressure-driven acceleration from a possible additional relative-velocity response. Net inflow into an existing atom is excluded. The parameter E is specified from orbital calibration, not obtained from an atomic sink rate. This comparison does not determine the cumulative front or remove the continuity conditions of the present study.

2 Reference frames and three possible transition criteria

Let $\mathbf{V}_c$ be the centre velocity and $\mathbf{\Omega}_b$ the body's rotation vector. The inertial and body-frame velocities are related by

$$\mathbf{u}_{\text{abs}} = \mathbf{V}_c + \mathbf{\Omega}_b \times \mathbf{r} + \mathbf{u}_{\text{rel}}. \tag{1}$$

The near-body condition under study is inward u_r with no tangential slip: $(u_{\text{rel}})_\phi = 0$. Thus radial motion relative to the body need not be a straight radial line in an inertial view. Adding the rotational coordinate term is a kinematic transformation and supplies no law of exchange with the body. Here "no tangential slip" means equality of tangential velocities only. The velocity equality does not specify a drag law or eliminate body–medium coupling. The inviscid continuum calculations below are conditional comparison models. Any proposed transport and conversion law must still satisfy continuity and the angular-momentum budget.

Author's phase-II condition. The proposed second phase distinguishes the central body from a test body carried by the local medium. In coordinates rotating with the central body, the stipulated common motion is

$$\mathbf{u}_{\text{rel}} = \mathbf{v}_{\text{rel}} = -U(r,t)\mathbf{e}_r, \qquad U > 0, \qquad \mathbf{w} = \mathbf{v} - \mathbf{u} = 0. \tag{2}$$

This is zero relative velocity between the test body and its local medium, while both move inward relative to the central body. In inertial coordinates both velocities include $\mathbf{V}_c + \mathbf{\Omega}_b \times \mathbf{r}$. Inflow from every direction and a body-independent radial acceleration are targets of the hypothesis. They are not established by this coordinate identity. For a radial speed field the radial component of the relative acceleration is $-\partial_t U + U\partial_r U$. A changing material boundary $R(t)$ requires its own trajectory derivative, as discussed below. The proposed radial representation $E = v_r^2 R^3$, with linear v_r, must be connected to the orbital representation while preserving the author's E. Neither equality of velocities nor a frame change supplies that connection. The criteria below are candidate diagnostics, not an already derived transition threshold. Pressure-only and passive-response closures elsewhere in this series are additional comparison models, not definitions of the author's mechanism.

At the equator of a spherical Kepler reference field,

$$\omega_K = \sqrt{E/r^3}, \qquad u_K = \sqrt{E/r}, \qquad v_{\text{eq}} = |\Omega_b| R_b. \tag{3}$$

Angular corotation, $\omega_K = |\Omega_b|$, gives

$$R_\Omega = (E/\Omega_b^2)^{1/3}. \tag{4}$$

Equating u_K at radius r to the surface speed instead gives

$$R_v = E/(\Omega_b^2 R_b^2) = R_\Omega^3/R_b^2. \tag{5}$$

It compares speeds at different locations. Equality of linear speeds at the same radius, $u_K = |\Omega_b| r$, returns R_Ω. Neither condition includes u_r. Neither determines where radial and tangential components are equal.

For an axisymmetric smooth field $u_\phi = r\Omega(r)\sin\theta$, the tangential speed vanishes on the rotation axis. A nonzero azimuthal speed on the exact axis has no single-valued smooth direction. Circular motion can occupy polar neighbourhoods, but cannot be extended to the axis with a fixed nonzero speed about that same axis. A poloidal component is required to transport material towards the equator. A field with $u_\theta = 0$ cannot do so.

3 Catalogue comparison: ten bodies, no individual fitting

3.1 Catalogue inputs and comparison conventions

Inputs are JPL orbital parameters and physical properties, supplemented by NASA rotation conventions [10, 11, 12]. They are catalogued estimates, not direct measurements of a medium. All radii below are measured from the centre, not heights above the surface. The formal $u_K(R_b)$ is an extrapolation of the spherical reference model to the equator, not a measured surface flow. The Sun rotates differentially. Its row uses 14.37 degrees per day at the equator. Signed retrograde periods are converted to speed magnitudes, so Venus's row is not a claim of vector corotation. Giant-planet rotation conventions and their variation limit the interpretation of small numerical differences.

Table 1: Formal surface speeds and angular corotation radii. Speeds are in km s^{-1}. Radii are in km.

Body	v_{eq}	$u_K(R_b)$	u_K/v_{eq}	R_Ω	R_Ω/R_b
Sun	2.0195	436.76	216.27	2.50663e+07	36.03
Mercury	0.0030263	3.0046	992.83	242884	99.521
Venus	0.001811	7.3266	4045.7	1.53655e+06	253.9
Earth	0.4651	7.9054	16.997	42164.2	6.6107
Mars	0.24073	3.5511	14.752	20427.6	6.0149
Jupiter	12.572	42.096	3.3484	160010	2.2381
Saturn	9.8712	25.087	2.5415	112239	1.8623
Uranus	2.5875	15.056	5.8188	82686.1	3.2351
Neptune	2.6829	16.614	6.1924	83507.8	3.3721
Moon	0.0046263	1.6795	363.04	88452.2	50.89

Every tabulated formal surface Kepler speed exceeds the equatorial material speed, but the ratio varies from about 2.54 to 4046. This is a useful quantified trend, not a universal speed ratio or a determination of a radial flux. The formal Moon value also requires caution because the Earth perturbs the large-radius domain. An isolated lunar calculation is not a valid description arbitrarily far from the Moon.

3.2 Rings and satellites constrain the proposed identification

For Earth, Eq. (4) gives 42164.17 km, consistent with the rounded geostationary radius. Its 0.17 km difference from a rounded catalogue value is not an independent precision measurement. A circular geostationary orbit is already an example of corotation without compulsory infall.

The familiar placement of Saturn's synchronous orbital radius within its B ring is reproduced here: the calculated 112239 km lies between the edges 91975 and 117507 km [13]. An alternative rotation period of 10 h 33 min 38 s yields 111567 km, also in that ring. The 0.60% shift is a rotation-convention sensitivity, not a statistical uncertainty interval. Uranus's narrow rings at roughly 41838–51149 km lie inside $R_\Omega = 82686$ km. The outer diffuse μ ring is centred near 97700 km [16]. Nearly circular orbital structures therefore exist on either side of the proposed dividing scale.

These observations do not measure u_r. They exclude the stronger claim that crossing R_Ω necessarily converts an orbiting, fully comoving body to radial-dominated motion. Small secular drifts, noncircular perturbations and body–medium slip remain distinct possibilities and need their own dynamical laws. The existence of rings is a constraint on a model, not a measurement of a sharp vacuum transition surface.

3.3 Measured slow drift on both sides of the corotation radius

Slow secular drifts are measured for two satellites, one on each side of R_Ω. Their direction is compared here with the position relative to R_Ω, without adjusting any parameter.

- The Moon orbits at about 384400 km, $9.117\,R_\Omega$ of the Earth. Lunar laser ranging shows that its distance increases, at about 3.8 cm per year [14].
- Phobos orbits at about 9380 km, $0.459\,R_\Omega$ of Mars ($R_\Omega = 20427.6$ km, Table 1). The analysis of its orbit from Earth-based astrometry, spacecraft imaging and spacecraft tracking yields a secular acceleration of its mean longitude, corresponding to a decreasing orbit [15].

Deimos, at about 23460 km or $1.148\,R_\Omega$ of Mars [12], is expected to recede, but no measured secular drift is used here. The inner satellites of Neptune inside its corotation radius are likewise expected to migrate inward. No measured drift is used here for these cases either. They are listed as expectations, not as evidence.

In both measured cases the sign of the slow radial drift agrees with the side of R_Ω: outward outside, inward inside. Inside R_Ω the orbital angular speed exceeds the spin of the central body, while outside it is smaller. The conventional description attributes both drifts to tidal exchange between the orbit and the rotation of the central body. Within the present framework the same observation is read as a constraint: R_Ω, which rings show is not a threshold of compulsory infall, is a dividing scale for the direction of slow radial drift in the measured cases. Whether that drift reflects relative circulation between an orbit and a medium corotating with the central body is a testable question. Its rates are not fitted here, and two measured cases do not establish a general law.

4 Continuity and the radial cubic tendency

4.1 Spherical stationary inflow

For an incompressible, source-free spherical flow at fixed time,

$$\nabla \cdot \mathbf{u} = 0 \quad \Longrightarrow \quad u_r = -Q/r^2. \tag{6}$$

Here $4\pi Q$ is the inward volume rate. With Keplerian equatorial rotation, the condition $|u_r| = u_K$ becomes

$$R_\chi = (Q^2/E)^{1/3}, \qquad |u_r|/u_K = (R_\chi/r)^{3/2}. \tag{7}$$

This establishes a calculable inward growth of radial dominance. Its amplitude requires Q, which the orbital invariant alone does not determine. Imposing $R_\chi = R_\Omega$ is the additional assumption $Q = E/|\Omega_b|$. For Uranus's inner ring near 41838 km this gives $|u_r|/u_K \simeq 2.78$. Such radial dominance is incompatible with a nearly circular, fully comoving ring trajectory. It is not rescued simply by a change of reference frame.

By contrast $u_r = -\sqrt{C_r}\, r^{-3/2}$ makes $u_r^2 r^3$ constant but has nonzero divergence in a stationary spherical domain. A stationary compressible version could have $\rho \propto r^{-1/2}$, since $\rho u_r r^2$ is constant. That explicitly changes the incompressibility assumption. Such a version is a classical comparison only. The author attributes incompressible fluid/gas properties to the proposed medium in the stated domain. These are process constraints in addition to the dimensional difference $[C_r] = L^5T^{-2}$ versus $[E] = L^3T^{-2}$. A proposed $C_r = E\ell^2$ would still require a physically specified length ℓ.

4.2 One flux normalization and two tangential profiles

Here we adopt the additional surface-inflow normalization $|u_r(R_b)| = \sqrt{E/R_b}$. In the notation used here this means

$$Q = \sqrt{E}\, R_b^{3/2}. \tag{8}$$

This is an assumption about flux amplitude, not a consequence of continuity and not a measurement of the surface medium. Keeping this same Q, the equatorial comparison with rigid corotation, $u_t = |\Omega_b| r$, gives

$$R_{\chi,\mathrm{solid}} = (Q/|\Omega_b|)^{1/3} = \sqrt{R_b R_\Omega}, \qquad \frac{|u_r|}{|\Omega_b| r} = \left(\frac{R_{\chi,\mathrm{solid}}}{r}\right)^3. \tag{9}$$

For the same flux compared instead with $u_K = \sqrt{E/r}$,

$$R_{\chi,K} = R_b, \qquad \frac{|u_r|}{u_K} = (R_b/r)^{3/2}. \tag{10}$$

The two radii therefore answer different questions about the tangential profile. They are not contradictory solutions of one fully specified flow. Both ratios are formed from centre-relative inertial components at the equator. In the body frame the tangential slip of rigid corotation is zero, so equality with that slip is a different, inapplicable criterion for nonzero radial inflow.

Table 2: One assumed flux normalization, two tangential comparison profiles. Radii in km from the centre. These are conditional calculations, not measured transitions.

Body	$R_b = R_{\chi,K}$	R_Ω	$R_{\chi,\mathrm{solid}}$	$\lvert u_r\rvert/u_K$ at $R_{\chi,\mathrm{solid}}$
Earth	6378.14	42164.17	16399.05	0.24256
Saturn	60268.00	112238.71	82245.99	0.62728

The Earth row uses the equatorial radius in the catalogue table. The difference from the earlier approximate 16390 km is an input-radius convention, not a new physical effect. For the rigid profile, a chosen ratio interval from 3 to 1/3 occupies radii $3^{-1/3}R_{\chi,\mathrm{solid}}$ to $3^{1/3}R_{\chi,\mathrm{solid}}$. Its width is $0.74889R_{\chi,\mathrm{solid}}$: about 12281 km for Earth and 61593 km for Saturn. The factors 3 and 1/3 define a reporting band. They have not been measured as physical transition endpoints. An interpolation from corotation to Keplerian rotation must still be supplied dynamically before this comparison predicts one physical transition. A free Q changes either equality radius.

4.3 A regular three-dimensional alternative

Let $P_2(\mu) = (3\mu^2 - 1)/2$, $\mu = \cos\theta$, and work in the shell $a \le r \le S$, outside the material body. A divergence-free family is

$$u_r = -A/r^2 - Br^{-3/2}P_2(\cos\theta), \tag{11}$$

$$u_\theta = (B/4)r^{-3/2}\sin\theta\cos\theta, \tag{12}$$

$$u_\phi = r\Omega(r)\sin\theta. \tag{13}$$

The meridional coefficient 1/4 is fixed by cancellation of the radial divergence. The associated streamfunction is

$$\psi = A\cos\theta - \frac{B}{2}\sqrt{r}\cos\theta\sin^2\theta. \tag{14}$$

The net inward flux is $4\pi A$. For $B > 0$, u_θ points towards the equator in both hemispheres, so the construction includes the meridional trend required by the hypothesis. At the axis and equator

this component is zero and regular. Trajectories of material bodies inherit that transport only if their velocity approaches this field. Continuity of the field alone does not prove entrainment.

Write $q(r) = B\sqrt{r}/A$ for $A > 0$. Inflow at every latitude requires

$$-1 < q(r) < 2. \tag{15}$$

The candidate radial invariant is

$$C_r(r,\theta) = \left(A/\sqrt{r} + BP_2\right)^2, \quad \left.\frac{\partial \ln C_r}{\partial \ln r}\right|_\theta = -\frac{1}{1+qP_2}. \tag{16}$$

Thus the cubic contribution can become locally important without becoming an exact global invariant. For $B > 0$ and all-direction inflow, its best fixed-latitude approach has $|\partial \ln C_r/\partial \ln r| > 1/3$. The pure B field has the exact radial exponent but zero net flux and both inflow and outflow regions.

4.4 A constructive local result: transport and the rotating boundary

The same meridional transport supplies more than a continuity correction. At a fixed time, writing $z = \cos^2\theta$, cancellation of the spatial transport $\mathbf{u}\cdot\nabla \ln C_r$ requires

$$f(z) = 1 - \frac{q}{2} + \frac{3}{2}(q-q^2)z + \frac{3}{2}q^2z^2 = 0. \tag{17}$$

For the positive-B branch, the first such points occur at $q = 5/3$, $z = 1/5$. For $5/3 < q < 2$ the two roots are

$$z_\pm = \frac{3(q-1) \pm \sqrt{3(3q-5)(q+1)}}{6q}. \tag{18}$$

They occur while the flow is still inward at every latitude. Thus a local compensation between radial and meridional transport is mathematically possible before outflow begins. At $q = 1.8$ the corresponding latitudes are approximately 13.2454° and 38.7598° in each hemisphere. This is a geometric prediction of the prescribed field, not a measured astronomical latitude. The temporal term $2\partial_t u_r/u_r$ must still be added. Zeros of f are not a material invariant along a future trajectory.

For a fixed spherical inner boundary, assume an initial $\Omega = \Omega(r)$ and inviscid axisymmetric evolution with scalar pressure. Let $\alpha = r\Omega'/\Omega$. The azimuthal equation gives

$$\partial_t\Omega = \frac{A\Omega}{r^3}\left[2 + \alpha - \frac{q}{6} + q\left(\alpha + \frac{5}{3}\right)P_2\right]. \tag{19}$$

With $B\Omega_b \neq 0$ and $\Omega(a) = \Omega_b$, a single latitude-independent surface spin derivative requires

$$\alpha(a) = -\frac{5}{3}, \qquad \dot{\Omega}_b = \frac{A\Omega_b}{6a^3}[2 - q(a)]. \tag{20}$$

For $A = 1$, $B = 0.9$, $a = 1$ and $\Omega_b = 1$, the required derivative is 0.1833333 in the illustrative units. This relates inflow geometry to the surface rotation condition. It does not supply the body's angular-momentum budget. The companion body–flow study treats a conditional boundary budget separately from the proposed cavity mechanism [6].

The recorded initial Euler example determines a normalized pressure π by

$$\nabla^2\pi = -\operatorname{tr}[(\nabla\mathbf{u})^2], \qquad \partial_r\pi = -[(\mathbf{u}\cdot\nabla)\mathbf{u}]_r \quad (r = a, S), \qquad \partial_t\mathbf{u} = -(\mathbf{u}\cdot\nabla)\mathbf{u} - \nabla\pi. \tag{21}$$

These boundary data fix $\partial_t u_r = 0$ on the two spheres. The source has spherical harmonics of degrees 0, 2 and 4. Its degree-4 coefficient in $\nabla\cdot D_u\mathbf{u}$ is $31B^2/(14r^5)$. A spherical $\Omega(r)$ cannot

cancel that coefficient. Accordingly this initial pressure construction is a defined dynamical test with a measurable mathematical residual, rather than an independently derived central acceleration. The full pressure solution and 39 checks will accompany the shared software release described in the Data and code availability section. Future boundary evolution, body response and exchange budgets are not closed by this initial construction.

5 Material acceleration and moving boundaries

5.1 Spatial velocity fields and cavity-wall motion

The relevant acceleration is $D_u\mathbf{u} = \partial_t\mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u}$. For steady spherical $u_r = -\sqrt{C_r}r^{-3/2}$ it is

$$(D_u\mathbf{u})_r = -\frac{3C_r}{2r^4}, \tag{22}$$

not $-E/r^2$. For a spherical incompressible flow $u_r = -Q(t)/r^2$ it is $-\dot{Q}/r^2 - 2Q^2/r^5$. The unsteady term can carry an inverse-square factor, but specifying it requires a boundary or evolution law. The radial exponent of a speed cannot be converted into an acceleration exponent by relabelling.

A useful neighbouring construction is the spherical Painlevé–Gullstrand river description [18]. Written with the orbital parameter E, its inward speed profile is $u_r = -\sqrt{2E/r}$. Direct differentiation in the stationary radial velocity-field calculation gives

$$u_r^2 r = 2E, \qquad u_r\partial_r u_r = -E/r^2. \tag{23}$$

This uses a radial exponent of $-1/2$, distinct from the cubic candidate's $-3/2$. The river description concerns a spacetime geometry. The displayed derivative compares radial profiles and is not a derivation of its full relativistic dynamics. Analogue-gravity models likewise relate prescribed physical systems to aspects of spacetime geometry [19]. Neither construction establishes the medium or the body-conversion mechanism hypothesized here. Their relevance is the explicit distinction between a speed profile, its material derivative and a physical interpretation.

For an inviscid spherical cavity with radius $a(t)$, density ρ and constant pressure difference $\Delta p > 0$, the Rayleigh equation [17] is

$$a\ddot{a} + \frac{3}{2}\dot{a}^2 = -\Delta p/\rho. \tag{24}$$

Released from rest at a_0, its first integral is

$$a^3\dot{a}^2 = \frac{2\Delta p}{3\rho}(a_0^3 - a^3). \tag{25}$$

The right side approaches a constant only late in collapse, $a \ll a_0$. At fixed time the exterior fluid still has $u_r = a^2\dot{a}/r^2$. The $a^{-3/2}$ wall-speed asymptote is therefore not the instantaneous spatial exponent of the entire exterior velocity field. This corrects the unqualified constant first-integral interpretation in the earlier radial-flow preprint [3]. Neither this analogy nor a successful integration of a separately imposed $\ddot{r} = -E/r^2$ supplies a constitutive body–flow mechanism.

More generally, if the initial inward wall speed is v_0, the same first integral contains $a_0^3v_0^2$ on the right-hand side of Eq. (25). For zero pressure difference and $v_0 \neq 0$, $a^3\dot{a}^2 = a_0^3v_0^2$ is exactly constant. Thus the moving-boundary cumulative result is retained: its interpretation depends on the initial state and pressure work. The wall radius $a(t)$, the spatial coordinate r of the medium and the position of a material test body are distinct variables.

5.2 A conditional spiral test of the transition geometry

A geometric pattern can be tested through its pitch rather than numerical resemblance alone. Define $\Phi = (1+\sqrt{5})/2$ and compare the inward logarithmic spiral $R = R_0 e^{-b\theta}$ with $b = 2\ln\Phi/\pi = 0.3063489625$, so each quarter turn contracts the radius by Φ. Its component ratio is constant: $\eta = |\dot{R}/(R\dot{\theta})| = b$. The mathematical definitions are given by Weisstein's Logarithmic Spiral and Golden Ratio. This fixed-pitch candidate alone does not produce a change from azimuth-dominated to radial-dominated motion.

A distinct, conditional construction assumes that one moving marker obeys both $R^3\dot{\theta}^2 = E$ and $R^3\dot{R}^2 = C$, with $\dot{R} < 0$ and $\dot{\theta} > 0$. Their simultaneous applicability is an additional assumption, not a demonstrated boundary–medium–body coupling. With $\ell = \sqrt{C/E}$,

$$\frac{\mathrm{d}R}{\mathrm{d}\theta} = -\ell, \qquad R = R_0 - \ell(\theta - \theta_0), \qquad \eta = \frac{\ell}{R}. \tag{26}$$

Thus component equality occurs at $R = \ell$, if the assumed laws remain valid there. This is not an independently predicted transition radius: ℓ and the applicability of both laws still need a closure. The construction approaches nearly circular motion at large R/ℓ but does not give an exactly circular outer region. It is a planar marker test, not the full three-dimensional flow or a material test-body trajectory.

For illustration only, set $E = R_0 = 1$ and $\ell = b$ to match the initial tangent of the two curves. The cumulative candidate reaches $\eta = 1$ at $R/R_0 = 0.306349$, whereas the golden candidate retains $\eta = 0.306349$. A finite-difference check of the analytic time solution agrees within 1.81×10^{-10} relatively in speed. This is a numerical control, not observational accuracy. The marker angular momentum $h = R^2\dot{\theta} = \sqrt{ER}$ changes as

$$\dot{h} = -\frac{\sqrt{EC}}{2R^2}. \tag{27}$$

Its transfer requires an explicit exchange law. The inviscid comparisons below do not supply the author's body–medium coupling. Neither pressure closure nor compatibility with near-body corotation is established by drawing the spiral. The exterior moving-boundary field $u_r(r,t) = R^2\dot{R}/r^2$ remains distinct from $\dot{R}$.

Successive Fibonacci ratios are mathematical approximants to Φ. The relative differences for 8/5 and 13/8 are 1.114562% and 0.430523%. These percentages do not measure the validity of a physical model. No celestial sample was fitted in these controls.

5.3 Unit coefficients and explicit coefficients

To expose the role of coefficients, let $x = R/R_{\mathrm{ref}}$ and $\tau = t/t_{\mathrm{ref}}$, with primes denoting $\mathrm{d}/\mathrm{d}\tau$. A unit-amplitude baseline and its general constant-amplitude extension are

$$x' = -x^{-p}, \qquad x\theta' = x^{-q}, \qquad \eta_0 = x^{-(p-q)}, \tag{28}$$

$$x' = -\alpha x^{-p}, \qquad x\theta' = \beta x^{-q}, \qquad \eta = \kappa x^{-\delta}, \quad \kappa = \alpha/\beta, \quad \delta = p - q, \tag{29}$$

where $\alpha, \beta > 0$ are amplitude coefficients in this normalized marker family. Here α is distinct from the rotational slope in Section 4.4. Equation (28) is the normalized form without explicit amplitude coefficients. It is a normalized baseline, not evidence that physical amplitudes universally equal one. In the cumulative–Kepler case, choosing $R_{\mathrm{ref}} = \sqrt{C/E}$ and $t_{\mathrm{ref}} = \sqrt{R_{\mathrm{ref}}^3/E}$ gives $\alpha = \beta = 1$. The physical length scale is retained in the units.

The general marker trajectory satisfies

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\kappa x^{1-\delta}, \qquad \begin{cases} x^{\delta} = x_0^{\delta} - \delta\kappa(\theta - \theta_0), & \delta \neq 0, \\ \ln(x/x_0) = -\kappa(\theta - \theta_0), & \delta = 0. \end{cases} \tag{30}$$

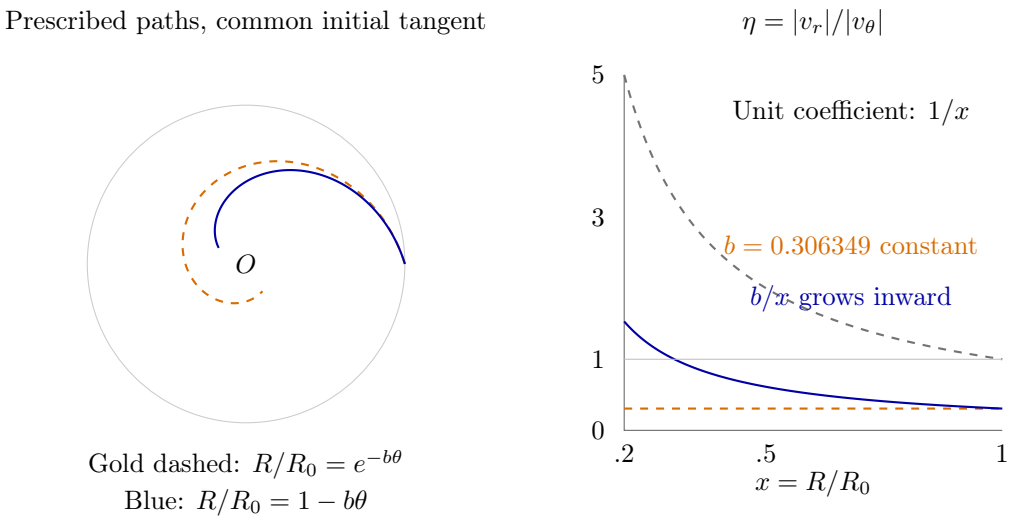


Figure 1: Conditional geometric comparison. The golden logarithmic spiral has constant component ratio $b = 2\ln\Phi/\pi$. The cumulative–Keplerian marker instead has $\eta = \ell/R$. Matching its initial tangent sets $\ell = bR_0$ only for illustration. The unit-coefficient curve is an additional prescribed case. None is selected by an independent observational fit. These are not proven temporal stages of one physical flow.

This exhausts the three exponent-sign classes within this planar, constant-coefficient power-law family. Inward radial dominance grows for $\delta > 0$, remains constant for $\delta = 0$, and decreases for $\delta < 0$. For $\delta \neq 0$, component equality occurs at $x = \kappa^{1/\delta}$ only if that radius is in the model's domain. For $\delta = 0$, equality occurs everywhere when $\kappa = 1$ and nowhere otherwise. A change of length units cannot remove the pitch κ from this logarithmic case.

Candidate	(p, q)	η with coefficients
Golden logarithmic control	$(1/2, 1/2)$	$b = 2\ln\Phi/\pi$
Cumulative plus Kepler	$(3/2, 1/2)$	κ/x
Cumulative plus conserved h	$(3/2, 1)$	$\kappa/\sqrt{x}$
Cumulative plus rigid rotation	$(3/2, -1)$	$\kappa/x^{5/2}$
Decreasing-radial control	$(1/2, 1)$	$\kappa\sqrt{x}$

The unit-coefficient forms follow by setting $\kappa = 1$. In the golden row that substitution changes the pitch and therefore ceases to be the golden candidate. A pure circle ($\alpha = 0$), a radial path ($\beta = 0$), and a stationary marker (both zero) are separate limiting cases. Parametrization by θ fails for a radial path. Reversing rotation reverses handedness, not the absolute component ratio.

The normalized acceleration components and angular-momentum change are

$$\widetilde{a}_R = -p\alpha^2 x^{-2p-1} - \beta^2 x^{-2q-1}, \tag{31}$$

$$\widetilde{a}_\theta = -\alpha\beta(1-q)x^{-p-q-1}, \tag{32}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(x^2\theta') = -\alpha\beta(1-q)x^{-p-q}. \tag{33}$$

Physical acceleration is $(R_{\mathrm{ref}}/t_{\mathrm{ref}}^2)\widetilde{a}$. Only $q = 1/2$ preserves the angular combination $R^3\dot{\theta}^2$ along this marker, and only $p = 3/2$ preserves $R^3\dot{R}^2$. The alternatives are comparison laws,

not simultaneous proofs of both invariants. With a prescribed physical E, β is fixed by $\beta = \sqrt{E}\, t_{\mathrm{ref}}/R_{\mathrm{ref}}^{3/2}$ in the Kepler case. It is not an extra independent fit parameter. Shared rescaling of α, β preserves geometry. To preserve physical E as well, that check rescales t_{ref} by the same factor.

Before evaluation, five exponent pairs and five coefficient pairs were fixed: $(\alpha, \beta) = (1,1), (1/2,1), (2,1), (b,1), (2,2)$. Twenty-five cases at four radii verify Eq. (30) with maximum relative inversion residual 1.23×10^{-15}. Independent finite differences check the normalized acceleration to 5.51×10^{-11} with the stated scaling. These are numerical checks of the family, not observational error percentages. A common coefficient change cannot be interpreted as unchanged physical rates unless the time normalization is accounted for.

For variable coefficients the local criterion becomes $\mathrm{d}\ln\eta/\mathrm{d}\ln x = \mathrm{d}\ln\kappa/\mathrm{d}\ln x - \delta$. Such coefficients require their own evolution law. They are not fitted freely to each body. Pressure closure, continuity, three-dimensional geometry and the identification of a marker with a material body remain separate tests. This classification covers the stated family, not all possible physical mechanisms. The stated exponent pairs, coefficient pairs and radius samples define the numerical control. Code availability is described below.

5.4 Pressure closure of the cumulative–Keplerian marker

The velocity construction above does not yet specify how the marker exchanges angular momentum. To test one possible closure, assume explicitly a pressure-only inviscid continuum with positive effective inertial density ρ, governed by $D_{\mathbf{u}}\mathbf{u} = -\nabla p/\rho$. This is an additional continuum assumption, not an inferred property of the hypothesized non-material medium. In cylindrical coordinates (s, φ, z), its azimuthal equation gives

$$D_{\mathbf{u}} h = -\frac{1}{\rho}\frac{\partial p}{\partial \varphi}, \qquad h = s u_\varphi. \tag{34}$$

Consequently an axisymmetric pressure cannot change a fluid parcel's h. If the planar cumulative–Keplerian marker is identified with such a parcel ($s = R$), its required $\dot{h} = -\sqrt{EC}/(2R^2)$ is incompatible with this particular closure for $E, C > 0$. This statement does not identify a moving boundary or a material body with that parcel.

Nonaxisymmetric pressure can redistribute rotation. However, at any fixed (s, z, t) a single-valued pressure obeys

$$\int_0^{2\pi} \rho D_{\mathbf{u}} h \,\mathrm{d}\varphi = 0. \tag{35}$$

Thus the same negative $D_{\mathbf{u}}h$ cannot hold at every azimuth of a complete ring with positive ρ under pressure alone. Negative changes along a selected streamline remain possible, with compensating positive changes elsewhere. This integral is not an assertion about the time average along that streamline.

In the dimensionless coefficient family, at $x = 1$ the unit-coefficient cumulative–Keplerian choice requires a pressure increment π after one full turn when $\widetilde{\rho} = 1 + 0.2\cos\varphi$. The illustrative choice $\alpha = 2\ln\Phi/\pi$, $\beta = 1$ requires $2\ln\Phi \simeq 0.962424$. Both increments are nonzero and therefore violate pressure periodicity. A 1024-point control for the valid pressure $\widetilde{p} = \sin\varphi$ has positive and negative local changes, with density-weighted integral 2.48×10^{-16} in magnitude. These are numerical checks of an analytical condition, not observational precision. The $q = 1$ member conserves parcel angular momentum and avoids this azimuthal obstruction. Its radial balance, continuity and connection to the outer law remain open.

Advection through a permeable body boundary can still deliver angular momentum even when parcels conserve it. That boundary budget, including changes of the body's inertia, is

treated in the companion study [6]. It does not supply the missing parcel-level change above merely by relabelling it as inflow. The present calculation isolates a requirement for a connected mechanism. It neither establishes nor rules out the broader proposed transition.

5.5 Cumulative inflow in a corotating frame

To retain the author's distinction between radial motion relative to the body and rotation relative to the surroundings, prescribe a marker at fixed colatitude ϑ, with the centre at rest and a constant body angular velocity Ω_b. These are explicit kinematic assumptions. For $C > 0$,

$$\dot{R} = -\sqrt{C}\, R^{-3/2}, \qquad \dot{\varphi} = \Omega_b, \tag{36}$$

$$R(t) = \left[R_0^{5/2} - \frac{5}{2}\sqrt{C}(t - t_0)\right]^{2/5}, \qquad \varphi(t) = \varphi_0 + \Omega_b(t - t_0). \tag{37}$$

The bracket must remain positive, and application stops at the physical inner boundary. No passage through $R = 0$ is inferred. With a rotation matrix $Q(t) = Q_z[\Omega_b(t - t_0)]$ and a fixed unit direction $\mathbf{n}_0$,

$$\mathbf{r} = Q(t)R(t)\mathbf{n}_0, \qquad \mathbf{r}_{\mathrm{rel}} = Q^{-1}\mathbf{r} = R(t)\mathbf{n}_0. \tag{38}$$

The corotating path is a radial ray, while the inertial path has an azimuthal component except on the rotation axis. Its velocity-component ratio is

$$\eta = \frac{\sqrt{C}}{|\Omega_b| \sin\vartheta\, R^{5/2}}, \qquad R_{\eta=1} = \left(\frac{C}{\Omega_b^2 \sin^2\vartheta}\right)^{1/5}. \tag{39}$$

This equality is relevant only within the domain where corotation and the radial law apply. At the poles, or for $\Omega_b = 0$, the inertial motion is already radial and there is no finite component-equality radius. The model keeps ϑ constant and therefore does not describe transport to the equator.

Using the earlier normalization gives both normalized forms:

$$x' = -x^{-3/2}, \qquad \varphi' = 1, \qquad \eta = \frac{x^{-5/2}}{\sin\vartheta}, \tag{40}$$

$$x' = -\alpha x^{-3/2}, \qquad \varphi' = \beta, \qquad \eta = \frac{\alpha}{|\beta| \sin\vartheta} x^{-5/2}, \tag{41}$$

Here α, β again denote normalized amplitudes, rather than the rotational slope. Specifically, $\alpha = \sqrt{C}\, t_{\mathrm{ref}}/R_{\mathrm{ref}}^{5/2}$ and $\beta = \Omega_b t_{\mathrm{ref}}$. Inward motion requires $\alpha > 0$. For positive Ω_b, choosing $R_{\mathrm{ref}} = (C/\Omega_b^2)^{1/5}$ and $t_{\mathrm{ref}} = 1/\Omega_b$ produces the unit-coefficient form. Thus coefficients encode physical scales even when absent from the normalized expression. For $\Omega_b \neq 0$, elimination of time yields $R^{5/2} = R_0^{5/2} - 5\sqrt{C}(\varphi - \varphi_0)/(2\Omega_b)$, which differs from the cumulative–Keplerian marker's linear $R(\varphi)$.

Radial motion in the rotating coordinates does not imply radial inertial acceleration. Direct differentiation gives

$$\mathbf{a} = Q\ddot{R}\mathbf{n}_0 + 2\boldsymbol{\Omega}_b \times (Q\dot{R}\mathbf{n}_0) + \boldsymbol{\Omega}_b \times (\boldsymbol{\Omega}_b \times \mathbf{r}), \qquad \ddot{R} = -\frac{3C}{2R^4}. \tag{42}$$

In particular, $h_z = \Omega_b R^2 \sin^2\vartheta$ and $\dot{h}_z = 2\Omega_b R\dot{R}\sin^2\vartheta$. If this marker is a fluid parcel, that change needs an angular-momentum mechanism of the kind tested in the preceding subsection. Coordinate conversion supplies none. Identification with a material body's $\mathbf{v}$, or with the medium velocity $\mathbf{u}$, and the evolution of $\mathbf{w} = \mathbf{v} - \mathbf{u}$, remain separate.

Matching the outer angular law to corotation gives the conditional radius $R_{\mathrm{match}} = (E/\Omega_b^2)^{1/3}$. At that radius, $\eta = \sqrt{C/E}/(R_{\mathrm{match}} \sin\vartheta)$: angular-rate equality does not by itself imply radial–azimuthal equality. With the fixed illustrative inputs $(E, C, \Omega_b) = (0.16, 0.09, 0.4)$, $R_{\mathrm{match}} = 1$, whereas $R_{\eta=1} = 0.891301$ at the equator and 1.023836 at $\vartheta = 45°$. The latter lies outside an inner-corotation domain $R \leq 1$, if that domain is imposed. The component ratio there is already greater than one at its edge. The parameter E remains fixed. Inside rigid corotation the angular expression $\Omega_b^2 R^3$ is not constant, so extending that outer representation inward would introduce an additional, unsatisfied condition. A connected transition law must specify how its representation changes.

A prescribed 18-state check covers two coefficient pairs, three colatitudes including the pole, and three radii. Rotating the positions back gives a maximum absolute discrepancy 3.89×10^{-16}. Independent finite differences of Cartesian positions agree with velocity and acceleration at maximum relative errors 1.62×10^{-9} and 4.69×10^{-7}, respectively. These are numerical verification errors, not observational uncertainties or evidence for the proposed medium. The coefficient pairs, colatitudes and radii define this synthetic control.

5.6 Continuity and flux relative to the moving interface

A boundary-speed law is not a through-body flux law. As an additional kinematic continuum assumption, take constant positive density and a finite spherical annulus outside $R(t)$, with constant $\mathbf{\Omega}_b$:

$$\mathbf{u} = \frac{A(t)}{r^3}\mathbf{r} + \mathbf{\Omega}_b \times \mathbf{r}, \qquad A = R^2\dot{R} = -\sqrt{CR}, \qquad \nabla \cdot \mathbf{u} = 0 \quad (r > 0). \tag{43}$$

At the moving inner sphere $u_r(R, t) = \dot{R}$, while elsewhere $u_r = R^2\dot{R}/r^2$. The cumulative boundary law and this instantaneous spatial dependence therefore coexist without being the same velocity profile. Rigid rotation is confined to the assumed annulus. This constructs a field compatible with continuity, not a pressure, energy or atom-conversion solution. The outer parameter E has not been changed. Its connection to this inner candidate remains to be derived.

Here $b(t)$ denotes an interface radius, distinct from the spiral pitch b in Section 5.2. For a spherical interface in the field domain, the inward volume flux relative to that interface and its advected axial angular momentum are, with the sign of Ω_b retained,

$$Q_{\mathrm{in}} = 4\pi(b^2\dot{b} - A), \qquad J_{\mathrm{in}} = \frac{2}{3}\rho\Omega_b b^2 Q_{\mathrm{in}}. \tag{44}$$

For the material boundary $b = R$, $\dot{b} = \dot{R}$, both fluxes vanish even though the boundary moves inward. At the same instant a fixed control sphere of that radius has $Q_{\mathrm{in}} = -4\pi A > 0$. Thus a contracting material interface cannot also be assigned a nonzero through-interface flux under the same boundary condition. A permeable body's own radius evolution and conversion budget must specify the normal relative velocity. The advected angular input in this conditional continuum model does not require viscous traction. Nonzero angular input alone still does not specify spin-rate change because body inertia can change. Normal pressure traction on a sphere has zero moment about its centre.

The corresponding material acceleration is

$$D_u\mathbf{u} = \left(\frac{\dot{A}}{r^3} - \frac{2A^2}{r^6}\right)\mathbf{r} + \frac{2A}{r^3}\mathbf{\Omega}_b \times \mathbf{r} + \mathbf{\Omega}_b \times (\mathbf{\Omega}_b \times \mathbf{r}), \qquad \dot{A} = \frac{C}{2R^2}. \tag{45}$$

At $r = R$, its first term has radial magnitude with sign $\ddot{R} = -3C/(2R^4)$, preserving the moving-boundary result. Off the axis, $D_u h_z = 2\Omega_b A s^2/r^3$ requires angular exchange, so continuity has

not removed the pressure-closure obstruction established above. For a test body with acceleration $\mathbf{a}_v$, the separate identity

$$\frac{d\mathbf{w}}{dt} = \mathbf{a}_v - D_u\mathbf{u} - (\mathbf{w}\cdot\nabla)\mathbf{u}, \qquad \mathbf{w} = \mathbf{v} - \mathbf{u}, \tag{46}$$

shows why universal body acceleration and approach to the flow cannot be inferred without a law for $\mathbf{a}_v$.

With the earlier dimensionless scales, $X = R/R_{\rm ref}$ and $\boldsymbol{\xi} = \mathbf{r}/R_{\rm ref}$ give $\mathbf{U} = -\alpha\sqrt{X}\boldsymbol{\xi}/|\boldsymbol{\xi}|^3 + \beta\mathbf{e}_z \times \boldsymbol{\xi}$ and $X' = -\alpha X^{-3/2}$. The unit-coefficient form sets $\alpha = \beta = 1$ by scale choice. A fixed 54-state finite-difference control gives maximum scaled divergence 6.41×10^{-10} and relative acceleration discrepancy 3.69×10^{-10}. Twelve interface cases verify the surface integrals to 1.78×10^{-15} absolute. These are numerical errors, not observational accuracy. These controls test the stated velocity field and interface assumptions.

6 HD142527: a local cubic tendency and an observational test

6.1 A result within a published velocity parameterization

Casassus et al. [20] describe resolved gas kinematics with a piecewise radial and azimuthal velocity prescription. Their section 3.1 explicitly cautions: "It is not a dynamically consistent model." We examine one branch of that prescription without fitting new constants. The symbol E denotes its fixed central orbital parameter. Using this notation does not turn the gas observations into a measurement of the proposed medium.

To avoid confusing the disk scale with the earlier poloidal coefficient A, write $L = 143\,\mathrm{au}$ and $b = 123\,\mathrm{au}$. Here b is a fixed disk-profile scale, distinct from the earlier pitch and moving-interface radius. On the published outer inflow branch $30 \le R < 130\,\mathrm{au}$, the speed can be written

$$|u_r| = k(L-R)^{7/2}R^{-1/2}, \qquad k = \frac{0.9\sqrt{2E/L}}{b^3}, \tag{47}$$

$$p(R) = -\frac{\mathrm{d}\ln|u_r|}{\mathrm{d}\ln R} = \frac{1}{2} + \frac{7}{2}\frac{R}{L-R}, \qquad C_r(R) = k^2R^2(L-R)^7. \tag{48}$$

Consequently $p = 3/2$ and $\mathrm{d}C_r/\mathrm{d}R = 0$ at $R_* = 2L/9 = 31.7778\,\mathrm{au}$. This is a local maximum of C_r within the parameterization. Normalizing by this maximum gives the following reproducible results, with E fixed throughout.

Table 3: Diagnostics of the Casassus et al. outer branch. These are calculated profile values, not independently measured radial velocities.

R (au)	$\eta = \lvert u_r\rvert/\lvert u_\phi\rvert$	p	$C_r/C_r(R_*)$
30	0.9743	1.4292	0.995872
31.7778	0.8914	1.5000	1.000000
35	0.7683	1.6343	0.987448
40	0.6225	1.8592	0.925534

Over 30–35 au the maximum relative departure from $C_r(R_*)$ is 1.256%. Extending the interval to 40 au raises it to 7.447%. These percentages describe profile variation about a selected local maximum. They are neither a residual against observations nor a confidence interval, and they do not establish $C_r = E$ or a universal cubic law.

Radial dominance and the cubic exponent remain different tests. In this prescription $u_\phi = (v_{\rm ff}^2 - u_r^2)^{1/2}$, with $v_{\rm ff}^2 = 2E(1/R - 1/L)$. Just outside 30 au, $\eta = 0.9743$, the separately prescribed

inner branch gives $\eta = 2.0647$ just inside. The formal outer-branch root $\eta = 1$ at 29.5026 au lies outside that branch and is therefore not a valid transition radius. Smoothing this join requires an evolution law rather than an unlabelled interpolation. The body-frame ratio $|u_r|/|u_\phi - \Omega_b R \sin\theta|$ also cannot be inferred without a specified stellar rotation and geometry.

6.2 Radial-scale sensitivity of the local tendency

Writing $x = R/L$ and $x_* = 2/9$ isolates the normalized shape:

$$F(x) = \frac{C_r(R)}{C_r(R_*)} = \left(\frac{x}{x_*}\right)^2 \left(\frac{1-x}{1-x_*}\right)^7. \tag{49}$$

A common distance rescaling $R, L, b \mapsto \lambda(R, L, b)$ leaves x, F and p unchanged. At fixed E, the absolute C_r instead scales as λ^2. Thus robustness of the normalized shape is not robustness of its absolute normalization. If E is estimated using the same distance, its covariance must be included separately. In general,

$$\mathrm{Var}(\ln x) = \mathrm{Var}(\ln R) + \mathrm{Var}(\ln L) - 2\,\mathrm{Cov}(\ln R, \ln L). \tag{50}$$

One shared distance factor cancels exactly. Uncertainty in relative geometry does not necessarily cancel. No unreported parameter errors are assigned here.

For $y = \ln(x/x_*)$, the curvature and local expansions are

$$\left.\frac{\mathrm{d}^2 \ln C_r}{\mathrm{d}y^2}\right|_0 = -\frac{18}{7}, \qquad F = 1 - \frac{9}{7}y^2 + O(y^3), \qquad p = \frac{3}{2} + \frac{9}{7}y + O(y^2). \tag{51}$$

The normalized invariant is therefore locally insensitive to first order, whereas the inferred exponent changes to first order. Illustrative changes of x by -5% and $+5\%$ give departures from the maximum of 0.3289% and 0.3136%, with $p = 1.4366$ and 1.5652 respectively. These are sensitivity scenarios, not measured uncertainty intervals.

The chosen tolerance also sets the reported radial extent. Within the valid outer branch, $F \geq 1 - \delta$ occupies 30–34.6452 au for $\delta = 1\%$, 30–35.0011 au for $\delta = 1.256\%$, and 30–38.4113 au for $\delta = 5\%$. Formal lower roots are outside that branch and are excluded. These bands quantify the profile's curvature. They are not confidence regions or measured transition boundaries. Local flatness near a smooth maximum alone cannot establish a universal physical law. The sensitivity follows from the normalized profile and its stated tolerances.

6.3 What the archived spectra do and do not constrain

We additionally inspected processed ^{12}CO and ^{13}CO cubes distributed with Garg et al. [21, 22]. Their spectral axes and spatial grids match. The velocity coordinate is line-of-sight velocity, not u_r. The native beams are approximately 0.3343×0.3033 and 0.3468×0.3061 arcsec, with different position angles. At the source's adopted distance 157.3 ± 1.2 pc these correspond to scales of roughly 48–55 au. A five-au radial interval is not independently spatially resolved by those beams. Model-based inference at smaller scales would require a forward model and its uncertainty assessment. The earlier prescription used a different distance scale. Physical radii must be harmonized before fitting.

Five spectra were extracted at a preliminary compact-continuum anchor and at four offsets of 0.343 arcsec. Their line profiles are not generally described by one unambiguous local velocity. In the ^{12}CO east and west spectra, changing the signal threshold from two to five times the estimated noise changes the centroids across ranges of 0.222 and 0.333 km s^{-1}, respectively.

Some other sampled profiles meet the edge of the available spectral window. The 0.084 km s^{-1} channel spacing is not a velocity uncertainty.

Garg et al. identify foreground-cloud and spatial-filtering effects over 3.0–5.7 km s^{-1} and describe their attempted correction as "highly uncertain" (Appendix A). The observed depression over 4.2–4.9 km s^{-1} falls within that affected interval. As a shape diagnostic, the mean intensity there divided by the mean in 3.4–4.1 and 5.0–5.7 km s^{-1} is 0.449 for ^{12}CO and 1.640 for ^{13}CO at the east sample. These native-beam ratios are not a common-resolution comparison, an opacity measurement or a detection significance. Different emitting regions and optical depths can matter. The central ^{13}CO peak has signal-to-noise about 4.12 and fails the adopted five-sigma peak criterion.

The exact continuum-subtraction and cloud-correction state of the supplied cubes has not been identified unambiguously from their metadata and headers. No such corrections were applied by assumption. Statistical noise, correlated resolution elements, masking, geometry and radiative-transfer systematics must all be separated from numerical errors. Uncertain input data can broaden an allowed model range, but do not by themselves validate a particular law.

An additional comparison convolves both cubes to a circular Gaussian beam of 0.400 arcsec FWHM. In east–north coordinates the kernel covariance is $\Sigma_{\mathrm{target}} - \Sigma_{\mathrm{native}}$, positive definite for both lines. Normalized kernels are applied to the intensity cubes, with the beam-area ratio converting Jy per native beam to Jy per target beam. Discrete covariance and point-source normalization checks have relative residuals below 1.8×10^{-9}. These check the convolution, not observational accuracy. At the east sample, the shape ratios become 0.424 ± 0.026 and 1.447 ± 0.160 for ^{12}CO and ^{13}CO, respectively. The quoted errors are conditional linearized noise estimates from 108 background-control positions, using the covariance of the two spectral-window averages rather than assuming independent channels. They are not full confidence intervals: control positions need not be independent or entirely free of emission, and calibration, processing and radiative-transfer uncertainties remain. The northern ^{13}CO ratio is withheld because its denominator is below three times the control dispersion. This test shows that the different Gaussian restoring beams alone do not explain the east-profile difference. It does not equalize residual dirty-beam effects or identify its unique physical cause. This comparison uses the cited processed cubes and the target beam specified above.

Further testing must assess rotation with a warp, radial flow and their combination through the same emission and instrumental response. This implements a test of η and p instead of interpreting a dip or a centroid alone as radial acceleration. The gas data provide an observational test case for flow geometry. They do not directly detect the non-material medium or its conversion into atoms.

6.4 A bounded test using full intensity-weighted moments

For a uniform thin ring in a known plane, with no vertical velocity, let $m_X = \langle X(v_{\mathrm{los}} - v_{\mathrm{sys}})\rangle$ and $m_Y = \langle Y(v_{\mathrm{los}} - v_{\mathrm{sys}})\rangle$. The averages use the full intensity, with coordinates centred and X along the line of nodes. Then

$$m_X = \frac{R u_\phi \sin i}{2}, \qquad m_Y = \frac{R u_r \sin i \cos i}{2}, \qquad \frac{u_r}{u_\phi} = \frac{m_Y}{\cos i \, m_X}. \tag{52}$$

A normalized spatial convolution with zero first moment preserves these mixed moments over the full aperture and spectral range. It adds its covariance to the spatial intensity covariance. This is an integration identity, not a claim of immunity to noise, incomplete apertures, spectral masks or absorption.

A 2048-point ring control with angular radius 0.202 arcsec, $i = 24°$, $u_r/u_\phi = -0.35$ and a common 0.400-arcsec Gaussian beam recovers that ratio to numerical precision when the geome-

try is supplied. A purely rotating ring with $i = 25.2787°$ and position angle shifted by $-17.7312°$ produces the same mixed-moment vector at the same assigned orbital parameter. The spatial covariances after convolution differ by 1.6447% when their Frobenius difference is normalized by the inflow case's covariance trace. This is a synthetic geometric difference, not an observed residual or significance. It identifies a separate intensity-shape constraint needed to distinguish the two cases. Non-axisymmetric emission and intrinsic eccentricity can affect it.

An additional closed Kepler ellipse with $e = 0.3$, apsidal angle 37° and inclination 24° has nonzero, sign-changing instantaneous radial velocity, but returns zero apparent radial ratio under residence-time weighting and constant tracer luminosity. In particular, the full-period average $\langle y\dot{y}\rangle = \langle \mathrm{d}(y^2)/\mathrm{d}t\rangle/2$ vanishes. Nonuniform emissivity or incomplete sampling requires a new control. These tests use ideal full moments rather than the observed masks. The five real spectra do not suffice to measure such moments. No radial velocity for HD142527 is inferred from this control.

6.5 An emissivity correction and its domain

For the same circular ring, let the positive azimuthal intensity weight be $1 + \epsilon \sin 2\phi$, $|\epsilon| < 1$, and write $\zeta = u_r/u_\phi$ for the signed ratio, distinct from $\eta = |\zeta|$. The simple moment ratio becomes

$$\zeta_{\mathrm{naive}} = \frac{\zeta + \epsilon/2}{1 + \epsilon\zeta/2}. \tag{53}$$

Thus $\epsilon = -0.7$ produces an apparent $\zeta = -0.35$ with zero radial motion. Conversely, $\epsilon = +0.7$ hides a true $\zeta = -0.35$ by making this indicator zero. These are assigned synthetic examples, not measured brightness patterns or flow ratios in HD142527.

A conditional correction is available from the same intensity distribution. Let S_{obs} be its centred spatial covariance, B the centred beam covariance and $\mathbf{m}$ the mixed position–velocity covariance. For a ring of fixed radius with azimuth-independent velocity components and known plane, $v_{\mathrm{los}} = \boldsymbol{\beta}^T(X, Y) + \text{constant}$, hence

$$\boldsymbol{\beta} = (S_{\mathrm{obs}} - B)^{-1}\mathbf{m}, \qquad \zeta = \cos i\, \frac{\beta_Y}{\beta_X}. \tag{54}$$

This follows from the weighted linear relation and requires an invertible intrinsic covariance, full spatial and spectral moments, and the stated convolution conditions. Four controls with $\epsilon = \pm 0.7$ and $\zeta = 0, -0.35$ recover the assigned ratio within 2×10^{-15}. This residual tests arithmetic in an exact model. It is not an accuracy claim for noisy observations or a new dynamical law.

The closed $e = 0.3$ ellipse explicitly fails the ring assumption. With the previous apsidal orientation, its unweighted naive indicator is zero but the ring correction returns -0.02335. With $\epsilon = -0.7$, the naive and ring-corrected indicators are -0.32161 and -0.03857, despite zero secular inflow. An eccentric orbital velocity template, its emissivity and sampling must therefore be treated together. The correction cannot be applied indiscriminately to the real disk. The assigned ring and ellipse examples delimit the correction's applicability.

6.6 Do the archived data support full moments?

A bounded audit of the existing archived ^{13}CO cube uses circular apertures of radii 0.5, 1.0 and 1.5 arcsec about the preliminary centre (zero-based row 156, column 176). All 110 channels are summed without positive clipping. For pixel area A_p and Gaussian beam area $A_b = \pi b_{\mathrm{maj}} b_{\mathrm{min}}/(4 \ln 2)$, the aperture spectrum is

$$F_k = \frac{A_p}{A_b} \sum_{j \in \mathcal{A}} (I_{jk} - b_k), \qquad J = \Delta v \sum_k F_k. \tag{55}$$

Here b_k is a channel-wise spatial-background median, not a subtraction of source continuum. The native beam is retained, $A_p/A_b = 0.019965831$ and $\Delta v = 0.084$ km s^{-1}. With background annulus 6–8 arcsec, $J = 0.37833$, 5.40610 and 15.26810 Jy km s^{-1}, respectively. Changing the background annulus among 6–8, 6–7 and 7–8 arcsec gives ranges of 0.588%, 0.162% and 0.129% relative to those values. These ranges quantify only that background choice, not total uncertainty.

The means of the first two and last two channels are compared with the root mean square from eight same-size control apertures on a 6-arcsec circle about the phase centre. At radius 1 arcsec, these ratios are 13.13 and 14.38. At 1.5 arcsec, 39.51 and 40.70. At 0.5 arcsec they are only 0.27 and 0.05. These ratios are empirical contrasts, not Gaussian significances: the controls need not be independent or representative of all image systematics. The channel centres span -1.000 to 8.156 km s^{-1} in LSRK.

Substantial edge brightness in the larger apertures and strong aperture dependence prevent assuming complete, isolated line moments from this product without further processing information. In particular, source continuum and line wings have not been separated. Edge brightness alone neither proves line truncation nor measures missing flux. A quiet edge in the smallest aperture does not establish completeness elsewhere. This is an audit of gas brightness, not a determination of u_r or of the proposed nonmaterial medium. The input archive is Garg's archived cubes. The original file MD5 was verified. The aperture sizes, spectral window and background choices above specify this audit.

The source provenance limits this interpretation further. In Garg et al. (2021), Fig. 1 uses peak-brightness products retaining continuum, whereas the caption of Fig. 2 describes continuum-subtracted ^{12}CO integrated products, with an exception for peak brightness. Section 2 also describes integrated maps retaining continuum. The archive's complete metadata identifies self-calibration but provides no explicit continuum-subtraction state for the archived ^{13}CO cube. The ^{12}CO figure caption cannot establish that state for the archived ^{13}CO file. Hence both routes are documented in the publication, but their file-level assignment remains unverified here. The preceding J values are brightness integrals of the supplied product, not established continuum-free line fluxes. No continuum correction or line-wing truncation is inferred from the edge diagnostic. This distinction follows from the cited publication and the archive metadata.

6.7 Observed spiral shapes available for a geometric test

Garg et al. [21], Section 3.2 and Figs. 4–6, trace brightness maxima beyond 210 au with radial cuts spaced by one degree. Appendix B, Table B1, gives the following coefficients for least-squares polynomial fits $r(\theta) = \sum_{j=0}^{3} a_j \theta^j$. These are transcribed source values, not parameters fitted here to obtain a golden pattern.

Tracer	Arm	a_0	a_1	a_2	a_3
^{12}CO	S1	546	-294	55.2	-3.31
^{12}CO	S2	-134	36.8	5.95	-0.99
^{12}CO	S3	-277	435	-185	26.5
^{12}CO	S4	37.9	-2.31	-0.42	0.64
^{13}CO	S1	496	-258	47.2	-2.75
^{13}CO	S4	-50.5	97.6	-34.1	3.78

The table preserves the original coefficient convention. It does not assign new physical units to coefficients whose units are not explicit there. The source gives dashes for ^{13}CO S2 and S3, not zero-valued fits.

These shapes trace emission features, not time-resolved particle trajectories. The paper also finds velocity perturbations and discusses vertical structure and wave propagation. Section 3.2

explicitly notes projection uncertainty. Its radial-distance comparison uses a wedge within ten degrees of the disc semimajor axis at position angle $-20°$. The reported inter-isotopologue radial offset is not a fitted-coefficient error bar.

For an angle in radians in a specified plane, the local pattern pitch would be

$$\tan\psi = \left|\frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}\theta}\right| = \left|\frac{a_1 + 2a_2\theta + 3a_3\theta^2}{a_0 + a_1\theta + a_2\theta^2 + a_3\theta^3}\right|. \tag{56}$$

A uniform radial-unit factor cancels, but angular units and domains do not. Table B1 does not supply coefficient uncertainties, covariance or angular validity intervals. The inspected captions do not explicitly resolve the polynomial's angular zero and unwrapping convention. Consequently no uncertainty-qualified comparison with the fixed golden pitch is reported here. All six curves are retained for a future convention- and geometry-controlled test. An isolated local-pitch coincidence would not establish a golden spiral, and a cubic polynomial fit to an image trace is distinct from the dynamical combination $u_r^2 R^3$. The coefficients above preserve the published six-curve set.

A version-specific check of the primary archive [22] on 6 October 2026 found five listed FITS products: three line cubes and two continuum images. Its complete manifest and descriptive metadata identify no separate trace-coordinate table, fitting script or angular-convention document. This is not a claim about unlinked resources or every FITS extension. The published six-curve set is retained, but the fixed golden-pitch test remains unevaluated for unresolved convention, domain and uncertainty inputs. No empirical percentage or significance is assigned, and this status is neither confirmation nor rejection of a golden pattern. The unresolved inputs are required before the geometric comparison can be evaluated.

7 Numerical checks

7.1 Numerical checks and results

The ten-body catalogue calculation records 30 algebraic substitutions with relative residuals below 2×10^{-14}. This checks implementation, not the measurement precision of the inputs. In the reported 39-check three-dimensional calculation, finite-difference divergence residuals are below 7.49×10^{-10} in the selected nondimensional test. For $A = 1$, $B = 0.9$, $a = 1$, $S = 4$, $q(S) = 1.8$ and inflow remains inward everywhere. Near the pole the cubic term exceeds the inverse-square term for $r > 1/B^2 = 1.234568$, while at $r = 4$ the logarithmic slope of C_r is -0.357143. This is a demonstrated tendency with a quantified departure from the proposed invariant.

8 Conclusions

Three transition criteria. We separate angular corotation, equality of radial and tangential speeds, and zero tangential slip as distinct conditions. We calculate formal corotation radii of ten bodies from catalogue data without adjusted constants. They reproduce the geostationary radius and Saturn's synchronous radius inside the B ring.

Corotation and the direction of slow drift. In both measured cases the drift agrees with the side of R_Ω: the Moon outside it recedes, Phobos inside it approaches. Conventional tidal descriptions share this sign, so it is a consistency check. Rings on both sides show that R_Ω is a dividing scale for drift, not a threshold of compulsory infall.

A three-dimensional inflow with a cubic tendency. We construct a divergence-free field combining net inward flux, an $r^{-3/2}$ radial contribution and meridional transport toward

the equator. All-direction inflow requires $-1 < q < 2$. The radial and meridional contributions to the spatial change of $u_r^2 r^3$ cancel locally at calculable latitudes before outflow begins. In the stated axisymmetric boundary example with $B\Omega_b \neq 0$, a latitude-independent time derivative of the surface spin requires $\alpha(a) = -5/3$.

Moving boundaries. We show that the cumulative wall law $R^3\dot{R}^2 = \text{const}$ coexists with a divergence-free exterior in which, at each instant, the flux is equal through all spheres. This refines the first-integral interpretation of the earlier radial-flow preprint [3].

HD142527. The published velocity profile has a local cubic tendency at $R_* = 2L/9 = 31.78$ au, with at most 1.256% variation of $u_r^2 R^3$ over 30–35 au within that profile. The archived CO data define the resolution, line-profile and moment requirements of an independent kinematic test.

Next step. A pressure and boundary evolution law, together with the body response of the companion studies [6, 23], is needed to turn these geometric results into one transition prediction for each body.

Declarations

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AI-based tools were used to assist with calculations, code development and manuscript preparation. The author takes full responsibility for the final content. No new experiment or observation is reported here.

Data and code availability

An earlier Ew2R3 visualization release is archived on Zenodo [7]. Calculations, input data and interactive visualization for this study are part of the Ew2R3 software release [8]. The programme website is Ew2R3.org.

The catalogue parameters and processed observational data are available from the sources cited in the relevant sections [10, 11, 12, 22]. The catalogues provide orbital and body properties, not velocities of a non-material medium. The original observations remain attributable to their cited providers.

References

[1] D. Kuzmenko, *The Gravitational Parameter $E = \omega^2 R^3$ as the Primary Constant of a Celestial System: Surface Gravity from Orbital Observations Without Masses or G*. ResearchGate preprint, doi:10.13140/RG.2.2.10828.37764.

[2] D. Kuzmenko, *The Orbital Invariant $E = \omega^2 a^3$ Across Scales: From the Hydrogen Atom to the Milky Way*. ResearchGate preprint, doi:10.13140/RG.2.2.24728.56327.

[3] D. Kuzmenko, *Radial Flow Interpretation of the Gravitational Invariant $E = \omega^2 R^3$: A Hydrodynamic Model for Free-Fall Acceleration Without Mass*. ResearchGate preprint, doi:10.13140/RG.2.2.33012.31364.

[4] D. Kuzmenko, *Kepler's Third Law and the Non-Material Medium: Cumulative Inflow, Atomic Cavities and a Common Origin of Weight and Inertia*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.34693.10721.

[5] D. Kuzmenko, *Kepler's Third Law in Multiple Systems: Component Orbital Parameters and Controlled Group Approximations*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.12882.72646.

[6] D. Kuzmenko, *Kepler's Third Law and Body–Flow Coupling: A Cavity Hypothesis and Conditional Compatibility Tests*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.19593.61282.

[7] D. Kuzmenko, *Ew2R3.org interactive visualization:* $E = \omega^2 R^3$, software, v0.1.0-rc.3 (2026). Zenodo. Release doi:10.5281/zenodo.21762453. Project doi:10.5281/zenodo.21757888. Website: Ew2R3.org.

[8] D. Kuzmenko, *Ew2R3 shared calculations and interactive visualization: matter, multiple systems, orbital-to-radial transition and body–flow coupling*. Computer software and data (2026) doi:10.5281/zenodo.23627938.

[9] K. Ilin and A. Morgulis, Inviscid instability of an incompressible flow between rotating porous cylinders to three-dimensional perturbations (2015, revised 2016). arXiv:1502.03600.

[10] NASA Jet Propulsion Laboratory, Solar System Dynamics: Astrodynamic Parameters. https://ssd.jpl.nasa.gov/astro_par.html. Input snapshot: October 2026.

[11] NASA Jet Propulsion Laboratory, Solar System Dynamics: Planetary Physical Parameters. https://ssd.jpl.nasa.gov/planets/phys_par.html. Input snapshot: October 2026.

[12] NASA NSSDCA, Planetary Fact Sheets, including Sun, Moon and planetary rotation entries. https://nssdc.gsfc.nasa.gov/planetary/planetfact.html.

[13] D.R. Williams, NASA NSSDCA, Saturnian Rings Fact Sheet (updated 19 April 2022). https://nssdc.gsfc.nasa.gov/planetary/factsheet/satringfact.html. Accessed 3 October 2026.

[14] J.G. Williams and D.H. Boggs, Secular tidal changes in lunar orbit and Earth rotation, *Celestial Mechanics and Dynamical Astronomy* 126, 89–129 (2016). doi:10.1007/s10569-016-9702-3.

[15] R.A. Jacobson, The orbits and masses of the Martian satellites and the libration of Phobos, *Astronomical Journal* 139, 668–679 (2010). doi:10.1088/0004-6256/139/2/668.

[16] NASA PDS Ring-Moon Systems Node, Vital Statistics for Uranus's Rings. https://pds-rings.seti.org/uranus/uranus_rings_table.html. Accessed 3 October 2026.

[17] Lord Rayleigh, On the pressure developed in a liquid during the collapse of a spherical cavity, *Philosophical Magazine* 34, 94–98 (1917). doi:10.1080/14786440808635681.

[18] A.J.S. Hamilton and J.P. Lisle, The river model of black holes, *American Journal of Physics* 76, 519–532 (2008). doi:10.1119/1.2830526. arXiv:gr-qc/0411060.

[19] C. Barceló, S. Liberati and M. Visser, Analogue Gravity, *Living Reviews in Relativity* 14, 3 (2011). doi:10.12942/lrr-2011-3.

[20] S. Casassus et al., Accretion Kinematics through the Warped Transition Disk in HD142527 from Resolved CO(6–5) Observations, *Astrophysical Journal* 811, 92 (2015). doi:10.1088/0004-637X/811/2/92.

[21] H. Garg et al., Non-Keplerian spirals, a gas-pressure dust trap, and an eccentric gas cavity in the circumbinary disc around HD 142527, *Monthly Notices of the Royal Astronomical Society* 504, 782–791 (2021). doi:10.1093/mnras/stab800.

[22] Garg et al., HD142527 processed data accompanying the 2021 study, Figshare, version 1. doi:10.6084/m9.figshare.13625144.v1.

[23] D. Kuzmenko, *Kepler's Third Law and the Flow of an Ideal Medium Around Atoms*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.33015.38562.

[24] D. Kuzmenko, *$E{=}\omega^2 R^3$: Kepler's Third Law as the Basis of a Unified Physical Picture: Results of the Ew2R3.org Programme and a Call for Independent Verification*. ResearchGate preprint (2026). doi:10.13140/RG.2.2.24393.30561.

A Continuity and streamline change

The radial divergence of the B term is $-(B/2)r^{-5/2}P_2$. The angular divergence of u_θ is $+(B/2)r^{-5/2}P_2$. The azimuthal divergence is zero. This proves incompressibility away from the excluded centre. At fixed time,

$$\mathbf{u}\cdot\nabla \ln C_r = \frac{A}{r^3}\left[1-\frac{3q^2\sin^2\theta\cos^2\theta}{2(1+qP_2)}\right]. \tag{57}$$

Its isolated zeros need not be persistent invariants. If $b_r = \partial_t u_r$, then $D_u \ln C_r = 2b_r/u_r + \mathbf{u}\cdot\nabla \ln C_r$. A spatial cancellation does not remove the time-dependent term. This distinction prevents a local zero from being interpreted as a universal transition surface.

B Reproduction and uncertainty

For a body with catalogue parameters E, R_b, T, recalculate $\Omega_b = 2\pi/|T|$, $q_b = \Omega_b^2 R_b^3/E$, $u_K(R_b)/v_{\rm eq} = q_b^{-1/2}$ and $R_\Omega/R_b = q_b^{-1/3}$. The table reports rounded results, without implying that catalogue inputs have independent Gaussian errors. Ring-source tables and different rotation conventions are kept separate. The three-dimensional field and derivative tests use the equations and assumptions stated in this study. All dimensionless flow coefficients are illustrative shared model parameters, not fits independently chosen for different planets.

The same assumed surface normalization applies to all ten catalogue rows. Both equality identities follow from that normalization. The comparison changes no measured input and supplies no observational determination of Q.